

MUZAMMIL AYAZ KABEER

Full Stack & Applied AI Engineer

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SUMMARY

Full stack and applied AI engineer with 6+ years of experience building production AI systems and backend services. I work on LLM agent orchestration, RAG pipelines, adaptive assessment engines, and the infrastructure that runs them. Most of my recent work has been 0 to 1: taking a rough product idea to something that serves real users daily.

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C++ (via node-addon-api)

AI & LLM: LangGraph, AutoGen/AG2, RAG pipelines, vector search, OpenAI embeddings, pgvector, prompt engineering (DSPy), structured LLM outputs, deepeval, LangFuse, Phoenix

Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Celery, TaskIQ, Redis, RabbitMQ, Gunicorn, Uvicorn

Frontend: React (Hooks, Redux), HTML/CSS, Streamlit, Electron

Databases: PostgreSQL, Aurora Serverless v2, MySQL, MongoDB, Redis, pgvector, ClickHouse

AWS: ECS, Fargate, EC2, ALB, ECR, S3, RDS, Secrets Manager, IAM, CloudWatch, VPC

Azure & Cloud: Azure OpenAI (model deployments, quota and region management), Azure AI Foundry, Azure AI Speech, Azure VMs, Cloudflare (AI Gateway, Workers, Pages), AssemblyAI

DevOps: Docker, GitHub Actions, CI/CD, SigNoz, OpenTelemetry, Linux, systemd, cron

EXPERIENCE

Senior Software Engineer, AI | iamneo

May 2024 - Present | Chennai, India

LSRW, AI language assessment lab (CEFR A1-C2, IELTS-aligned)

- Built the autopilot engine that decides what each student practices next. A day-one diagnostic places them on the CEFR ladder, then two loops run independently: session length escalates after consecutive strong sessions and drops back on regression, while level promotion needs consistency across a rolling 5-7 session window. All four skills must clear threshold separately, so a strong reader cannot carry a weak speaker.
- Solved semantic repetition in the question generator with an approach I call Negative RAG. Accepted questions are embedded and stored; new candidates get rejected on cosine similarity, and rejected ideas go back into the prompt as negative context through a LangGraph retry edge. Three rejections in a row trigger a saturation check and a topic pivot. No student sees the same question twice.
- Wrote the bundling algorithm that assembles 24-question IELTS bundles (12 reading, 10 listening, 1 writing task, 1 speaking cue card) at a single CEFR level, with LCG-based distribution for non-repeating delivery. Kept content generation and autopilot selection fully separate.
- Built the speech and scoring pipeline: Azure AI Speech TTS with CEFR-tiered speech rate control, AssemblyAI transcription, and tiered evaluation from rule-based keys through BERT similarity to LLM rubric grading.
- Deployed on AWS ECS/Fargate behind an ALB with Terraform, Aurora PostgreSQL Serverless v2, Secrets Manager, and FastAPI plus TaskIQ workers. Serves hundreds of student sessions daily.

NeoEval, automated code grading and academic integrity

- Built a persona classifier that sorts each code submission into one of four bands: non-attempt, wrong approach, right approach with mistakes, or correct and clean. Test-case pass/fail says whether code works; this says how the student got there.

- Designed the LLM output schema behind it (is_genuine_attempt, approach, execution, why, confidence) plus the composite-to-band scoring. Every classification is auditable, and low-confidence cases route to human review instead of publishing automatically.
- Set up the evaluation harness with deepeval and generated a synthetic C++ dataset with deliberate persona-subtype variation, which caught misclassification on ambiguous submissions before rollout. Later extended the model to SQL and JDBC questions with repair-effort scoring.
- Designed an integrity score on top of ClickHouse keystroke telemetry, turning typing dynamics into features for gradient-boosted models (LightGBM, XGBoost) that flag sessions as original, copied, or malpractice.

AI Copilot and platform services

- Built an agent-orchestrated AI Copilot with intent-based routing across specialist agents (explainer, summarizer, technical expert) and a separate sub-graph for test-case validation and debugging. Shipped beginner and expert modes using Socratic prompting.
- Built a token-metered AI usage system with wallet management, real-time deduction, and role-based access control, so AI consumption stays fair and auditable across thousands of exam sessions.
- Developed async AI backend services on FastAPI, Celery, Redis, and Docker, with GitHub Actions pipelines for zero-downtime deploys.

Full Stack Developer | VisAI Labs

May 2021 - Apr 2024 | Chennai, India

- Built a Linux desktop application in Electron, React, and Node.js for high-precision 3D object dimensioning, deployed across multiple manufacturing sites.
- Integrated C++ measurement modules into Node.js through node-addon-api for sub-millisecond hardware communication and real-time data capture.
- Designed Docker-based microservices for edge devices, covering automated backups and over-the-air software updates to industrial machines.
- Built RabbitMQ pipelines moving measurement data from distributed machines to central reporting, cutting data lag by roughly 60%.

Programmer Analyst Trainee | Cognizant

Aug 2020 - Apr 2021 | Chennai, India

- Worked on enterprise database-driven applications and picked up the fundamentals of large-scale architecture, team development workflow, and production code standards.

EDUCATION

B.Tech, Mechatronics Engineering | SRM Institute of Science and Technology | 2020 | 79%

ADDITIONAL

Languages: English, Tamil, Hindi, Urdu

Interests: System design, chess, anime, cooking
